Steering Gear Box

Service and Diagnostic Manual

Class 8 Hydraulic Power Steering System

Part Number: SGB-6500

Revision: C

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Classification: Dealer Restricted

Supersedes: All previous revisions

Section 1: General Description

The SGB-6500 Steering Gear Box is a hydraulic-assist recirculating ball type steering gear that provides precise directional control for Class 8 on-highway and vocational vehicles. The gear box converts rotational input from the steering column into linear thrust at the pitman arm.

Section 2: Component Identification

The SGB-6500 consists of seven sub-component groups described in Table 2-1.

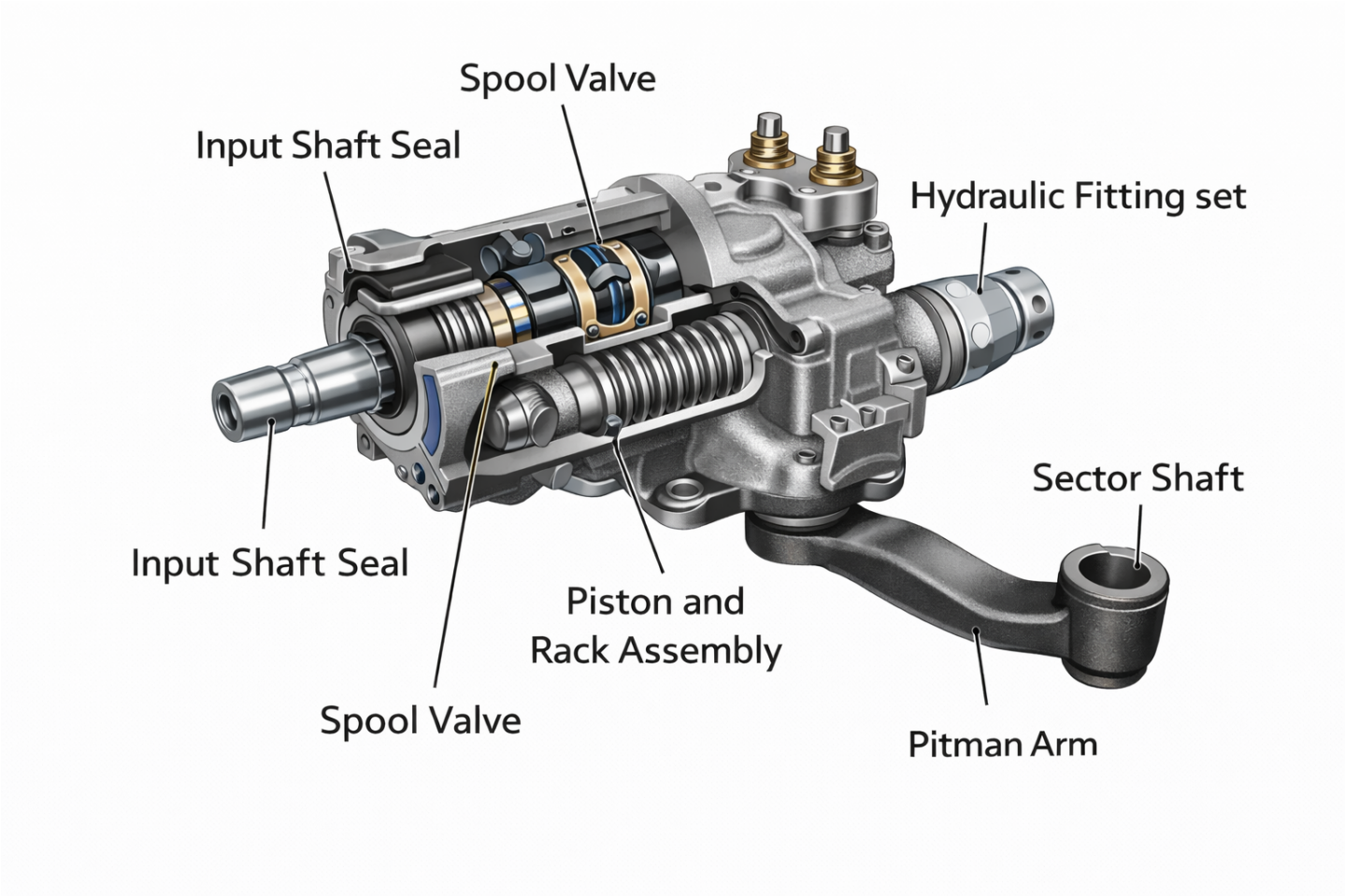

Figure 2-1: SGB-6500 Steering Gear Box - Component Layout

Table 2-1: Sub-Component Identification

Item	Sub-Component	Description
1	Input Shaft Seal	Double-lip seal with dust excluder at the steering column interface. Prevents hydraulic fluid leakage externally and contaminant ingress internally.
2	Spool Valve	Torsion bar-actuated directional control valve. Directs pressurized fluid to the appropriate side of the piston based on steering wheel input torque and direction.
3	Piston and Rack Assembly	Hydraulic actuator converting fluid pressure into linear steering force. Contains recirculating ball mechanism for low-friction translation of rotary to linear motion.
4	Sector Shaft	Output shaft transmitting steering torque from the piston rack to the external pitman arm through a tapered gear mesh. Factory-adjusted for zero backlash at center.
5	Housing Casting	Ductile iron enclosure forming the hydraulic cylinder bore for the piston and mounting structure for all internal components. Rated to 2,200 PSI working pressure.
6	Hydraulic Fitting Set	O-ring face seal fittings for pressure supply and return line connections from the power steering pump. Includes port plugs and diagnostic test port fittings.
7	Pitman Arm	Forged steel output lever connecting the sector shaft to the vehicle steering linkage via the drag link. Splined interface with castle nut retention at 250 ft-lbs.

Section 3: Specifications

The following specifications apply to all SGB-6500 configurations.

Type	Recirculating ball, hydraulic power assist
Ratio	17.5:1 overall (variable)
Maximum System Pressure	2,200 PSI (relief valve setting)
Normal Operating Pressure (straight ahead)	150 - 250 PSI
Normal Operating Pressure (full lock)	1,600 - 1,900 PSI
Fluid Type	TITAN MOTORS-approved ATF or power steering fluid
Fluid Capacity (gear box only)	3.2 quarts
Input Shaft Seal Type	Double-lip with dust excluder
Sector Shaft End-play	0.000 - 0.002 inches (adjusted)
Pitman Arm Nut Torque	250 ft-lbs + cotter pin
Housing Mounting Bolt Torque	225 ft-lbs
Assembly Weight	85 lbs

Section 4: Installation and Removal

Support the front axle assembly. Disconnect the drag link from the pitman arm using a ball joint separator - do not use a pickle fork. Disconnect pressure and return hydraulic lines and cap all open lines. Disconnect the steering column intermediate shaft. Remove frame mounting bolts and lower the gear box. During installation, apply thread-locking compound to mounting bolts and torque to 225 ft-lbs. Use new O-ring seals at each hydraulic fitting. Bleed air by turning lock-to-lock several times with engine running.

Section 5: Theory of Operation

Hydraulic Assist Mechanism

When the driver turns the steering wheel, the input shaft rotates a torsion bar-actuated spool valve within the gear box. The spool valve directs pressurized fluid to one side of the piston and rack assembly while returning fluid from the opposite side. The torsion bar provides proportional feedback: harder turns open the spool valve further.

Pressure Relief Function

A spring-loaded poppet relief valve in the housing casting opens when system pressure exceeds 2,200 PSI, bypassing pump output to the return circuit. This protects the gear box and pump during conditions like hitting a curb or holding against a mechanical stop.

Section 6: Troubleshooting and Diagnostics

6.1 Steering Wander at Highway Speed

The need for constant small corrections to maintain straight-line travel is most commonly caused by excessive clearance in the sector shaft mesh with the piston and rack assembly. The gear teeth wear over high mileage, creating a dead band at center position where input shaft rotation does not produce corresponding pitman arm movement. The driver perceives this as vagueness at the straight-ahead position. Measure rotational free play at the input shaft with the drag link disconnected. Acceptable play is 2 to 4 degrees. If play exceeds 6 degrees and adjustment cannot bring it within specification, the sector shaft and piston and rack assembly are worn beyond serviceable limits.

6.2 Increased Steering Effort

A gradual increase in steering effort, particularly during low-speed maneuvering, that the driver may not immediately recognize because they have adapted incrementally, can indicate that the relief valve spring in the housing casting has weakened or the poppet seat has eroded. The relief valve opens at pressures below its intended 2,200 PSI setting, silently bypassing hydraulic assist force. This symptom is easily confused with a worn power steering pump or low fluid level. Check relief valve crack pressure by monitoring system pressure at the hydraulic fitting set diagnostic port during a full-lock turn - pressure should reach 2,100 to 2,200 PSI before relief.

6.3 Fluid Leak at Steering Column

Fluid seepage past a degraded input shaft seal may initially appear as barely perceptible dampness on the steering column below the floor seal, often mistaken for condensation. The seepage rate increases as the seal lip wears, eventually becoming a visible drip on the frame rail below the steering gear. Additionally, the degraded seal may allow air to be drawn into the low-pressure side of the spool valve during rapid steering input, producing a characteristic groaning or moaning sound distinct from a pump whine. If foaming is visible in the fluid reservoir after the noise occurs, air ingestion through the input shaft seal is confirmed.

6.4 Leak at Hydraulic Connections

Fluid weeping at the pressure or return line connections indicates deterioration of the O-ring seals in the hydraulic fitting set. The O-rings harden over time from heat exposure and chemical interaction with power steering fluid additives. Overtorqued fittings can also deform the sealing surface. Use UV-fluorescent dye and a UV lamp to trace the exact leak origin point.

6.5 Pitman Arm Looseness

Clunking noises during direction changes originating from the pitman arm area indicate that the pitman arm splined interface has developed play on the sector shaft. This occurs when the castle nut has loosened or the cotter pin has sheared, allowing the pitman arm to shift on the splines under steering load reversal. Inspect the castle nut torque and cotter pin condition. If the splines show visible fretting or wear marks, both the pitman arm and sector shaft require replacement.

Section 7: Preventive Maintenance

At each B-service, check fluid level and condition. Inspect input shaft seal area and all hydraulic fitting connections for weeping. At each C-service, measure steering free play at the steering wheel rim with the engine running - maximum allowable is 2 inches. Check pitman arm castle nut torque and cotter pin condition. At 300,000 miles, proactively replace the input shaft seal and hydraulic fitting set O-rings.